

Resisting Social Pressure in the Household Using Mobile Money: Experimental Evidence on Microenterprise Investment in Uganda[†]

By EMMA RILEY*

I examine whether changing the form of disbursement of a microfinance loan enables female microfinance borrowers to overcome intra-household sharing pressure and grow their businesses. Using a field experiment with 3,000 borrowers in Uganda, I compare the disbursement of a loan as cash to disbursement onto a digital account. After 8 months, women who received their microfinance loan on the digital account had 11 percent higher (US\$70) business capital and 15 percent higher (US\$18) profits compared to those who received their loan as cash. Impacts were greatest for women who experienced pressure to share money with others in the household at baseline. (JEL C93, D13, G51, J16, O12, O16)

Most women in developing countries are self-employed, running their own microenterprises (International Labour Organization 2020). However, their businesses often remain small. Microfinance was held up as a solution to promote growth by enabling lending to poor women. However, on average, microfinance has had little effect on business outcomes (Banerjee, Karlan, and Zinman 2015) and seems to only work for some (Banerjee et al. 2019; Meager 2019, 2022). Additionally, while evidence has highlighted high rates of return to capital for men, returns to capital for women are puzzlingly small (De Mel, McKenzie, and Woodruff 2008, 2009). What constrains women from investing in and growing their businesses?

One possible constraint on female enterprise growth, despite access to business loans or grants, is intra-household sharing pressure, with family diverting funds away from the woman's business and the woman underinvesting as a result (Jakiela and Ozier 2016; Squires 2018). In cases where women are not subject to family

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sharing pressure, either because they can hide money or they own the only household business, women are able to expand their businesses in response to grants or loans (Fiala 2017; Bernhardt et al. 2019). This suggests that finding ways to help female entrepreneurs circumvent intra-household sharing pressure could improve their business performance. Crucially, sharing pressure has been shown to be different for cash on one's person versus money kept in other forms (Platteau 2000; Anderson and Baland 2002). This indicates that moving away from cash toward digital forms of money may bypass sharing pressure.

In this paper, I examine whether changing the form that a microfinance loan is disbursed in, from cash to a private digital account, changes women's business investment. I test whether sharing pressure is the channel by which changes in the form of a loan influence its use, as opposed to alternative channels such as self-control or saving constraints.

I conduct a randomized controlled trial of 3,000 female existing microenterprise owners in Kampala, Uganda. Clients of the microfinance NGO BRAC Uganda applying for a loan were individually randomized equally into two treatments or a control group. In the Mobile Account treatment, a mobile money account labeled as for the woman's business was provided to the woman, but the loan continued to be disbursed as cash. In the Mobile Disbursement treatment, women received a loan that was disbursed directly onto the mobile money account rather than disbursed as cash. A control group continued to receive their microfinance loan as cash and were not given a new mobile money account.¹ All study arms had to go to the branch to have their loan disbursed and continued to repay their loan as cash at weekly group meetings within their community.

Eight months after the microfinance loan was disbursed, the Mobile Disbursement treatment led to a 15 percent (US\$18) increase in business profits and an 11 percent (US\$70) increase in the value of business capital compared to the control group. Both total household income and household consumption are significantly higher in the Mobile Disbursement group, suggesting improving women's business performance improves outcomes at the household level. I do not find any effects from the Mobile Account treatment, suggesting that the loan must be deposited onto the mobile money account for there to be beneficial impacts.

I use transaction records provided by the mobile money operator to gain an in-depth picture of how the accounts were used. Those assigned to the Mobile Disbursement treatment held significant balances on the account immediately after the loan was disbursed: on average, US\$120, approximately 33 percent of the loan value, during the first 7 days of account ownership, and US\$60 during the first 15 days after loan disbursement, 50 percent of women's baseline savings value. The Mobile Disbursement group drew the loan down over a three-month period by making multiple withdrawals and using the mobile money account as a way to safely store the loan. Neither the Mobile Disbursement nor Mobile Account group used the accounts for regular deposits of their own money, with only 13 percent of either treatment group ever making a single deposit of their own onto the account, and of

¹ Note that 97 percent of the women had used mobile money before, so the treatments principally look at the impact of designating a mobile money account for business use.

very small amounts (median US\$7.20).² This is aligned with other research studies that have found that depositing money directly into an account does not necessarily cause people to make deposits themselves (Somville and Vandewalle 2018; Field et al. 2021).

I examine the potential mechanisms by which the Mobile Disbursement treatment had an impact on the women's businesses by looking at whether the treatment targeted primarily self-control difficulties, helped women to resist family pressure to share money, or just provided a safe place to store money. The treatment effects are driven by those who experienced the most pressure at baseline to share money with family: this group saw a 25 percent (US\$30) increase in business profits from receiving their loan on a mobile money account and a 26 percent (US\$180) increase in business capital, compared to the control group. I do not see heterogeneous impacts of either treatment by an index of self-control difficulty or evidence that the women were saving constrained.

Next, I use a machine learning approach to examine the characteristics of those experiencing the greatest treatment effects, finding it is women with characteristics linked to high sharing pressure, as well as more profitable businesses, who benefit most from the Mobile Disbursement treatment. The sharing pressure appears to come from the spouse, with married women with a spouse present at home benefiting the most from the Mobile Disbursement treatment. Looking at the use of the loan immediately after it was disbursed, I see less of the loan being shared, and more saved, for women assigned to the Mobile Disbursement treatment. Crucially, money saved by the Mobile Disbursement group remained on the mobile money account, private and earmarked for the business. The transaction data confirm that women in the Mobile Disbursement group who experienced sharing pressure stored more of their loan on the mobile money account for longer than those who did not. Taken together, this suggests the Mobile Disbursement treatment worked primarily by providing a way to store the loan in a manner less subject to sharing pressure from the spouse, so that it could be used for investment into the businesses when needed in the weeks after the loan disbursement.³

The lack of impact of the Mobile Account treatment also opens the question of why this group didn't simply imitate the Mobile Disbursement group and deposit the entire loan directly onto the mobile money account that they received at disbursement.⁴ It is possible that learning is a constraint, with women needing to experience the benefits of depositing the loan on a mobile money account before doing so themselves. Indeed, women in the Mobile Disbursement group are more likely to report that they'd want to receive a subsequent loan on a mobile money account. However, this does not translate into behavior: women in the Mobile Disbursement

² Mobile money accounts have been shown to be an effective way to save for business expenditures, school fees, health expenses, agricultural inputs, and unexpected shocks (Dizon 2017; Bastian et al. 2018; Blumenstock, Callen, and Ghani 2018; Habyarimana and Jack 2024; Lipscomb and Schechter 2018; Batista and Vicente 2020).

³ I rule out a series of alternative hypotheses that could explain the impact of the Mobile Disbursement treatment, including whether the increase in profits is just a redistribution of income within the household, with other household businesses losing out (Bernhardt et al. 2019); whether the woman experienced any backlash from her spouse; whether improved record keeping could explain my findings; whether there were changes in the woman's risk-sharing network; and whether her loan repayment behavior changed.

⁴ Other studies have found similar dependence on the form capital is initially given in, particularly for women (Fafchamps et al. 2014).

group do not deposit subsequent loans, and women in the Mobile Account group do not begin to deposit their loan themselves over time if they observe other women receiving their loan this way. This suggests that learning is not the answer. I argue instead that it is most likely due to behavioral default effects, where the loan is sticky in the form it starts off in,⁵ combined with procrastination, small transaction costs, and potentially underestimation of the benefits of depositing the loan onto the mobile money account.

My paper contributes to the literature on how to make microfinance effective. The literature has found little growth on average in enterprises after receiving microfinance loans (Banerjee, Karlan, and Zinman 2015) and found that when cash grants are given to female-owned enterprises, little growth in profits results (De Mel, McKenzie, and Woodruff 2008, 2009). However, studies have found that grants and loans for female-owned businesses can lead to increases in profits if women are able to hide money from their spouse (Fiala 2017), live in households with no other members who have businesses (Bernhardt et al. 2019), or receive a grant in-kind (Fafchamps et al. 2014). This paper expands this literature by showing that if female entrepreneurs are given their loan in a digital form that keeps it separate and earmarked for their business, they are able to invest more of the loan into their business and experience high returns on their business investment.

Second, I contribute to the literature on inefficiency within the household by showing that sharing pressure is a first-order constraint on female entrepreneurs using microfinance loans to expand their businesses. A large number of field and experimental studies have shown that individuals are willing to take costly actions and change their behavior in order to hide or maintain control over their income (Ashraf 2009; Baland, Guirkinger, and Mali 2011; Schaner 2015; Jakiela and Ozier 2016; Boltz, Marazyan, and Villar 2019; Castilla 2019; Carranza et al. 2021). I likewise see that women employ costly ways to spend the loan themselves rather than hold cash and feel pressured to give it to others: the control group spends much of the loan on household durable goods, and in focus groups women described struggling to hold the loan as cash for their businesses due to requests for money from their spouse and so preferred to spend it on household goods that they valued. Using a variant of the game utilized in Almås et al. (2018), I find over 50 percent of women are willing to give up US\$8 for the spouse in order to control US\$2 themselves. I show that reducing the need for women to rely on these costly hiding strategies by providing a labeled account for them to store their loan raised income for the household as a whole.

Lastly, my paper is the first to show that the default way in which a loan is disbursed matters for investment. Default effects have been shown to be large across a wide range of domains (Choi et al. 2004; Chetty et al. 2014b). A growing literature in developing countries has shown that defaulting savings into accounts results in significantly higher savings, despite often minuscule costs to people of depositing the money into the account themselves (Brune et al. 2016, 2017; Blumenstock, Callen, and Ghani 2018; Somville and Vandewalle 2018; Brune, Chyn, and Kerwin

⁵Default effects have been shown to be extremely powerful drivers of saving behavior, with high degrees of inertia to keep money in whatever form it starts off in, regardless of the benefits of switching (Brune et al. 2017; Blumenstock, Callen, and Ghani 2018; Somville and Vandewalle 2018; Brune, Chyn, and Kerwin 2021).

2021). Direct deposit of wages into a bank account has been shown to be necessary for the bank account to result in higher control of the money for women (Field et al. 2021). My research extends this literature to credit by showing that the default manner in which a loan is disbursed determines the extent to which it is invested in a business.

The rest of this paper is organized as follows: Section I contains the conceptual framework. Section II discusses the interventions and experiment design. Section III describes the data and the empirical specification. Section IV provides the main finding. Section V discusses mechanisms, and Section VI examines why the Mobile Account group did not deposit their loan themselves onto the mobile money account. Section VII discusses alternative explanations, and Section VIII concludes.

I. Conceptual Framework

The treatments could change the extent to which the loan is subject to sharing pressure within the household through a number of channels (see online Appendix A for a model). I classify these channels broadly as providing a private saving device, mental accounting (labeling) effects, commitment effects, and default effects.

Private Saving Device: The mobile money account provides a private and safe storage device for keeping the loan on and saving business income, negating the need to hold cash. This decreased visibility may reduce unplanned expenditures on personal items or pressure to give money to others. At baseline, 20 percent of the sample reported carrying some savings as cash, despite having bank accounts and ROSCAs. Since mobile money accounts provide no interest and there are fees for withdrawing money (though not for depositing money), the accounts effectively have a negative interest rate. The mobile money account may represent an in-between point of saving flexibility: it is more accessible than a bank account or ROSCA but less accessible than cash.

Mental Accounting: Evidence suggests that simply labeling something as a saving account can increase savings (Thaler 1985, 1999), and previous studies have indeed found that a separate, labeled mobile money account can increase saving for the labeled purpose (Dizon 2017; Habyarimana and Jack 2024; Lipscomb and Schechter 2018). Money in this saving account is viewed as being unavailable for day-to-day spending. This therefore helps people to resist the temptation to spend the money on other things. It may also help them to resist family pressure, as they can utilize the label to argue the money has been allocated for another purpose already.

Soft Commitment Device: Providing the microfinance loan on a mobile money account may act as a soft commitment device compared to giving the loan as cash, as it requires a trip to a mobile money agent to actively withdraw money before spending it. This contrasts with cash, which is easy to spend instantly. This would not necessarily be the case if paying for goods with mobile money was common, but less than 1 percent of mobile money users have used it to pay for goods (Intermedia 2016). The commitment features of the mobile money account may help to resist

the pressure to give money to others. Moreover, while sending money to others is a feature of mobile money accounts, it still requires more steps than to simply hand them some cash. It also requires the receiver to withdraw the money from an agent the other end and to pay a fee. This may therefore be enough to dissuade others that it is worth asking for money from the women, and so reduce social pressure to share.

Default Effects: A common theme across these mechanisms is the default difference between the treatments. The Mobile Account treatment requires active deposit of funds onto the account. The Mobile Disbursement treatment, however, automatically provides a safe place labeled for the business to store the loans until the money is actively withdrawn. Prior literature has shown these default effects are an important predictor of savings (Brune et al. 2017; Blumenstock, Callen, and Ghani 2018; Somville and Vandewalle 2018). Changing the default way that money is held may therefore help women overcome both self-control and social sharing pressure challenges.

II. Background and Experiment Design

A. *Mobile Money*

Mobile money use was already high when this study took place: 51 percent of the population used mobile money services in Uganda in 2017 (Demirguc-Kunt et al. 2017), and over 40 percent of users were women. In Kampala, women's use of mobile money services was considerably higher (93 percent), and 87 percent of women owned their own mobile phone (Intermedia 2016). Hence, this study took place in a context of widespread use and availability of mobile money.

Mobile money services operate via a simple SMS message interface on a SIM card to allow the transfer and storage of up to US\$1,000. The account is PIN protected and so can only be accessed by the owner. As such, they do not require a smartphone or Internet access. Withdrawal and deposit of money take place using networks of mobile money agents, who are found at extremely high density in a city like Kampala and operate extended hours. There are fees for withdrawing or transferring funds but not for depositing money.

B. *BRAC Microfinance Loans*

BRAC is one of the largest NGOs in the world and one of the largest providers of financial services to the poor in Uganda. BRAC offers microfinance loans of between US\$70 and US\$1,000 for expanding a small enterprise to women only. Owning an existing enterprise is a prerequisite for obtaining a microfinance loan. Loan durations vary between 20 and 40 weeks, with the interest rate set at 13 percent for the 20-week loan and 25 percent for the 40-week loan. Women apply for loans individually but must form a group of between 8 and 30 women from their community. Since the groups are formed within a community, the women generally know each other well. Groups are not formally liable for repayment of their members' loans, and women each have a guarantor from outside the group who is meant to repay the loan if a woman defaults.

Women meet weekly with the other group members to repay their loans in cash. This weekly meeting is held with a credit officer in the women's community. The women receive their loan as cash⁶ and make repayments also in cash, as is usual for microfinance loans in sub-Saharan Africa (World Bank Group 2020). When women receive a new loan, they must complete the required paperwork at the BRAC branch. On average, the branches are 1.49 miles (a 40-minute walk) from the women's businesses.

C. Study Population

The study location is Kampala, Uganda, chosen as it has both a high prevalence of microfinance borrowing and high mobile money penetration. The study took place in six of the microfinance branches of BRAC Uganda in the Kampala and Entebbe areas. The study population was composed of any microfinance client applying for a new loan with BRAC (whether as a first-time borrower or a repeat loan) who owns a mobile phone of her own.⁷ This sample of women is therefore representative of female microfinance clients in Kampala and likely similar to other urban populations of microentrepreneurs.

D. Treatment Arms

The study involved two treatments and a control group. Women in all arms were already approved for their loans and had to go to the branch to learn about their treatment status and receive their loan.⁸ All other aspects of the BRAC microfinance loan product remained the same, and women continued to repay their loan as cash at weekly group meetings.

Mobile Account: Women in this arm were offered a new SIM card and mobile money account while waiting at the branch to collect their loan. The account was described as specifically for their business, and suggestions were made like taking payments and saving for their business, but no formal restrictions were placed on use. If the woman accepted, she received immediate one-on-one assistance from an enumerator in setting up the mobile money account, including setting a secure PIN number and information on how to withdraw and deposit money and how to check the balance. Women in this group received their microfinance loan as cash as usual, irrespective of whether they accepted the mobile money account or not.

Mobile Disbursement: Women in this arm were offered the same business mobile money account and information as in the Mobile Account treatment, but, additionally, their microfinance loan was deposited directly into this mobile money account

⁶At the time of the study, BRAC was not a bank and so could not open deposit accounts for the women. Bank account ownership in the sample was 28 percent, and a bank account was not an option for receiving the loan in.

⁷Only six women were excluded from taking part in this study because they did not have their own mobile phone.

⁸Women were told that BRAC was testing a new method of disbursing loans, and so some women would be offered mobile money accounts and disbursement of their loan on a mobile money account and others not. BRAC frequently piloted changes in their products and services, so this is not unusual.

by BRAC. An extra amount was deposited to the mobile money account with the loan to cover the fee of approximately 1 percent of the loan amount for withdrawing the money from an agent, so as not to disadvantage women receiving the loan this way.⁹ Since women had to pay the fee to withdraw the loan, they would end up with the same amount of cash after withdrawing it as if they had received their loan as cash. This was fully explained so as to maximize take-up. Women could refuse either the disbursement or the SIM card, permitting partial compliance if she wanted the SIM card but not the disbursement. If a woman refused the disbursement of the loan onto the SIM card, she received her loan as cash.

Control: Women received the loan as cash at the branch. Nothing was changed from the existing loan disbursement process of BRAC.

E. Randomization

The study included 3,000 female microentrepreneurs, assigned using an individual stratified randomization as follows: 1,000 to the control group, 1,000 to the Mobile Account arm, and 1,000 to the Mobile Disbursement arm.

Randomization took place weekly in batches of 250 women determined by the timing of requesting a new loan. Within each batch, all women accepted for a loan were individually randomized into the treatment or control groups. Randomization continued for approximately 5 months until the sample size of 3,000 was achieved. Lists of treatment assignments from the randomization were sent to the BRAC branches weekly, and only women who had completed the baseline survey and had been assigned a treatment could have a loan disbursed to them.

Since assignment to treatment took place over a number of months and women applied for loans on different schedules, within the same group, women would be assigned to different treatment arms at different time points or not be in the study at all.¹⁰ I use this to examine potential spillovers and learning effects (see Section VIA).

The randomization was stratified by five variables at baseline: a dummy variable capturing present bias from a multiple price list incentivized game (Harrison et al. 2002), a dummy variable capturing if the woman switched above the median in a willingness to pay to hide money from the spouse game (Almås et al. 2018) (see Section IIIA for more details on the incentivized games), a dummy variable capturing if the client is a first-time borrower with BRAC, the microfinance branch, and a dummy variable for above-median business profit. The first two variables capture that women with self-control difficulties or who experienced sharing pressure might benefit more from the Mobile Disbursement treatment. I stratified by first-time borrower in case there were systematic differences between new and established

⁹This amount would cover five withdrawals of approximately one-fifth of the loan. Fees are set amounts based on tiers of withdrawals, rather than being a fixed percentage. The additional fee amount was only deposited to the mobile money card alongside the loan if the woman agreed to receive her loan on the mobile money account. If the woman refused the mobile disbursement but still received the mobile money account, there would be zero balance on this mobile money account, as in the Mobile Account treatment.

¹⁰On average, groups had 20 members and 5 were in the study. Due to the relatively large number of microfinance groups in the study (476), there are 48 groups that only had 1 study participant.

entrepreneurs, by branch to ensure a balanced load of mobile money disbursements, and by profit, as Fafchamps et al. (2014) showed heterogeneous effects of loans for women by profitability.

III. Data and Empirical Strategy

I use five sources of data for the analysis, three of which were self-reported by the women and two of which are administrative data. I conducted a baseline and endline survey, both of which were carried out by BRAC's Research and Evaluation Division, which is a separate legal entity to BRAC microfinance and carries out independent evaluations of BRAC programs in collaboration with academic researchers (Riley 2024).

A baseline survey was conducted on all women applying for a new loan at the six BRAC microfinance branches. Baseline surveys were conducted between January and June 2017 before randomization and assignment to treatment group occurred. The baseline survey focused on self-reported questions around business, household, and financial outcomes. Behavior games with incentives were also played (see Section IIIA).

The endline survey covered similar topics to the baseline survey. It began in October 2017 and ran until January 2018. This is approximately 8 months after the loan disbursement and was chosen so that the majority of women who had 40-week loans were still repaying them at endline, helping to reduce attrition.

Focus groups were conducted with a sample of 16 women from 3 different microfinance groups during September 2018 to obtain qualitative information on how women used the mobile money account and how they felt it affected their businesses. There were eight women from the Mobile Disbursement treatment, five from the Mobile Account treatment, and three from the control group. Though this is a small sample, the focus groups give a deeper understanding into the mechanisms by which the treatments had an impact.

Additionally, I use two sources of admin data. First, I obtained transaction records from MTN Uganda of all the mobile money transactions between January 2017 and January 2018 made using the mobile money accounts provided to clients as part of the study (all respondents gave their consent for this). The data include the type of transaction (including transfer, payment, cash-in, cash-out), account numbers for whom the transaction was from and to, date and time, amount, fee, and balance on the account. The transaction records are available for both treatment groups but not the control group. Second, I also obtained BRAC administrative data on loan repayment behavior.

A. Behavioral Games

Incentivized games were played at baseline (and endline) to elicit time preferences and willingness to pay to hide money from the spouse. I stratify the randomization by these measures in order to ensure balance and increase my power to look at heterogeneity by these measures.

The time preference games used were standard multiple price lists (Ashraf, Karlan, and Yin 2006; Andersen et al. 2008). Individuals were asked to choose between a

fixed monetary reward in one period and various larger rewards in a later period. The periods were either today and two weeks' or two weeks' and four weeks' time. The near payment was fixed at US\$2, and the far payment varies between US\$1.8 and US\$8. One in five respondents was randomly chosen to be paid one of her choices from this game at the specified time period.

The propensity to pay to hide money from the spouse game has been used as a measure of women's empowerment in the literature (Ashraf 2009; Fiala 2017; Almås et al. 2018). I expand upon Fiala (2017) by conducting a variant of the Almås et al. (2018) game with multiple choices between whether the woman or her spouse receives set amounts of money the next day. Women had to make a series of eight choices between receiving a fixed amount of money themselves (US\$2) or having their spouse receive varying amounts of money between US\$1.8 and US\$8. The choices are illustrated in online Appendix Table A1. One in five respondents was randomly chosen to be paid one of her choices from this game to either herself or her spouse tomorrow. Tomorrow was chosen to be the payment date to remove effects of strong present bias and to allow the enumeration team time to contact and find the spouse if necessary.

B. Balance Test and Baseline Characteristics

I confirm the validity of my randomization by performing a balance test (online Appendix Table A2). A joint orthogonality test cannot reject that all the characteristics are jointly zero for the Mobile Account treatment (p -value 0.38), the Mobile Disbursement treatment (p -value 0.88), or either treatment (p -value 0.15).

A few sample characteristics are worth highlighting: 80 percent of the women had completed primary school, and 15 percent completed secondary school. On average, they were 35 years old with 3 other household members. Two-thirds of them were married, and 20 percent had a job in addition to their business.

Looking at the game behavior, 20 percent of the women displayed hyperbolic preferences, which is similar to other studies (Ashraf, Karlan, and Yin 2006). Almost 60 percent of them switched above the median in the hiding money game, meaning they were willing to give up US\$6 in order to retain control over US\$2 rather than have their spouse be given it. This large amount of hiding is similar to that found in other studies (Fiala 2017; Almås et al. 2018) and suggests inefficiency in the household.

The average loan was US\$380 (see online Appendix Figure A1), and half the loans were for 40 weeks. Women reported making US\$120 a month in their businesses. Eighty percent of women had previously taken a loan from BRAC. Women had been running their businesses for six years on average. The women's business capital was predominantly in inventory, which made up 80 percent of the total capital stock on average. As online Appendix Figure A2 shows, over 55 percent of the businesses were selling items without any direct value being added to them, such as selling foods in a market, selling clothes, or operating small shops. Food stalls in particular have a highly perishable form of stock that would require regular purchases. In general, the women operated working capital-type businesses, rather than fixed capital-based businesses, and so needed to frequently purchase new stock with money saved from the previous period's sales. In terms of more

capital-intensive businesses, 8 percent operated a hairdressing salon, 6 percent did farming, and 4 percent made and repaired clothes.

Married women lived in a household where their spouse had a business 57 percent of the time. Nearly 90 percent had savings, and these averaged US\$130. Ninety-seven percent of women reported already having used mobile money before, and the nearest mobile money agent was less than 5 minutes from their home. The households earned on average US\$290 a month, so the woman's business brought in 40 percent of household income. They spent US\$240 a month and owned US\$620 in household assets on average.

C. Take-Up

Women were free to accept or reject the assigned treatments, in which case they would receive their loan as cash as usual. This means take-up of the interventions is important to examine.

Take-up is high: 94 percent of the individuals assigned to Mobile Account received a mobile money account, and 71 percent of those assigned to Mobile Disbursement received both a mobile money account and the loan disbursed onto the mobile money account (Figure 1).¹¹

Partial compliance was also possible in the Mobile Disbursement arm, with women receiving a mobile money account and their loan as cash (they were assigned to receive Mobile Disbursement and got Mobile Account). The reasons for this were both refusal of the disbursement of the loan onto the mobile money account (5 percent) but also external problems completing mobile disbursement, such as power cuts or networks outages (10 percent). In these cases, the mobile money account was still offered if desired, resulting in 14 percent of the Mobile Disbursement group receiving a SIM card but the loan as cash. Lastly, 15 percent of women assigned to Mobile Disbursement refused the entire treatment (SIM card and mobile disbursement) and just received their loan as cash. Refusal of the SIM card is higher in the Mobile Disbursement arm than the Mobile Account arm (15 percent versus 6 percent). This is possibly because when the SIM card was offered alone, most women were happy to accept it, whereas when women were offered the loan on the SIM card, if they said no to the mobile disbursement they also said no to the SIM card alone.

D. Attrition

Attrition was low: 90 percent of the sample were found and resurveyed for end-line. Attrition rates of approximately 10 percent are common in mobile populations such as this urban sample. Of the 10 percent who were not resurveyed, 25 refused to be surveyed and 292 couldn't be found. Moreover, attrition was not differential

¹¹I look at correlates of take-up in online Appendix Table A3. Married women in particular are less likely to take up the Mobile Disbursement treatment.

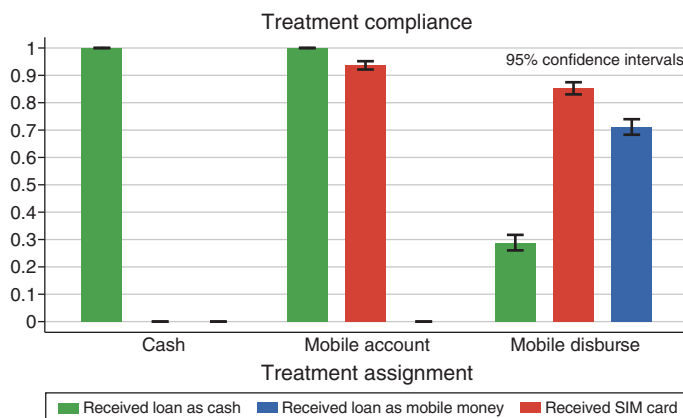


FIGURE 1. TAKE-UP BY TREATMENT ARM

Notes: Means and 95 percent confidence interval; of the 29 percent of women in the Mobile Disbursement group who received their loan as cash, 10 percent had accepted the loan on the mobile money account but were unable to receive it due to random network outages, and 19 percent refused the loan as mobile money and received it as cash.

by treatment status: online Appendix Table A4 shows no significant differences in attrition rates across treatment arms.¹²

E. Empirical Strategy

Survey Data.—I follow McKenzie (2012) and estimate intent-to-treat (ITT) effects using an ANCOVA specification:

$$(1) \quad Y_{is1} = \alpha_0 + \alpha_1 T_{1i} + \alpha_2 T_{2i} + \alpha_s + Y_{is0} + \epsilon_{is},$$

where Y_{is1} is the outcome of interest for individual i in stratum s , T_1 is a dummy variable for assignment to the Mobile Account treatment, T_2 is a dummy variable for assignment to the Mobile Disbursement treatment, α_s is a set of randomization strata dummies (Bruhn and McKenzie 2009), Y_{is0} the baseline value of the outcome (if measured at baseline, otherwise excluded), and ϵ_{is} is a random error term. Hypothesis tests will use standard errors that allow for heteroskedasticity. The parameters of interest are α_1 and α_2 , which give the effect of assignment to the Mobile Account and Mobile Disbursement groups respectively on the outcome variable. I also test whether the treatments have a significantly different effect from each other ($\alpha_1 = \alpha_2$).

All outcomes were prespecified in a pre-analysis plan, and any departures from this are noted.¹³ As I am considering multiple primary and secondary outcome measures, I adjust the p -values of the coefficients of interest for multiple statistical

¹²Correlates of attrition are shown in online Appendix Table A5. Older women, those in larger households, those with larger loans, women who were impatient in the incentivized game, and women who reported their family takes their money are less likely to be surveyed at follow-up.

¹³The pre-analysis plan is available at <https://www.socialscisearch.org/versions/32364/docs/version/document>, and an amendment at <https://www.socialscisearch.org/versions/32363/docs/version/document>.

inference by calculating sharpened q -values that control for the false discovery rate (FDR) within an outcome family. Rather than prespecifying a single q , I report the minimum q -value at which each hypothesis is rejected, following Anderson (2008) and Benjamini, Krieger, and Yekutieli (2006). For some summary measures of outcome families and when looking at heterogeneous effects, I group several related variables into index variables following Anderson (2008).

Mobile Money Operator Data.—The data from the Mobile Money Operator are only available for the two treatment groups that I gave mobile money accounts to. I estimate ITT effects for the administrative data using a regression specification of the form

$$(2) \quad Y_i = \alpha_0 + \alpha_2 T_{2i} + \alpha_s + \epsilon_i,$$

where Y_i is the outcome of interest, T_2 the Mobile Disbursement treatment dummy, α_s is a set of randomization strata dummies, and ϵ a random error, for individual i .

IV. Results

A. Primary Business Outcomes

I first examine self-reported profits, total savings, and the value of enterprise capital (defined as the value of business assets and inventory), results for which are shown in Table 1. I find a positive and significant effect on both profits and business capital for the Mobile Disbursement treatment. Both of these results also remain after a multiple testing correction is applied. Those in the Mobile Disbursement treatment experienced a 15 percent (US\$18) increase in their self-reported profits and an 11 percent increase in the value of their business capital (US\$70) compared to the control group.¹⁴ The increase in business capital from the Mobile Disbursement treatment of US\$70 is approximately 18 percent of the mean loan value of US\$380. The Mobile Disbursement treatment therefore seems to be increasing profits and increasing capital investment into the business.

There are no effects of the Mobile Disbursement treatment on the total amount of saving when measured eight months after the loan was disbursed.¹⁵ I find no significant effects of the Mobile Account treatment on any of the three outcomes, and I am able to reject equality of the treatment effects for the Mobile Account and Mobile Disbursement treatments for both business profits and business capital but not savings. The values at the bottom of Table 1 report the mean value of the outcome in the control at baseline and endline. The control group do not see any changes in

¹⁴In online Appendix Table A6, I examine the different components of profit: monthly and weekly sales and calculated monthly and weekly profits. I observe similar patterns with the Mobile Disbursement treatment increasing significantly all measures of sales and profits.

¹⁵In online Appendix Table A7, I look at alternative measures of saving, finding that savings on a mobile money account increase a small amount from both treatments, though this seems to be at the expense of other forms of saving.

TABLE 1—TREATMENT EFFECTS ON WOMAN’S BUSINESS PROFITS, SAVINGS, AND BUSINESS CAPITAL

	Profit (1)	Savings (2)	Capital (3)
<i>Mobile Account</i>	2.88 (3.61) [0.43] {1.00}	0.93 (9.54) [0.92] {1.00}	13.48 (24.18) [0.58] {1.00}
<i>Mobile Disburse</i>	17.61 (3.54) [0.00] {0.00}	8.46 (10.23) [0.41] {0.75}	69.21 (23.87) [0.00] {0.01}
Observations	2,639	2,639	2,639
R^2	0.44	0.35	0.51
Control mean	109.81	155.33	678.28
Control mean baseline	116.62	134.32	655.90
p -value $MA = MD$	0.00	0.43	0.01

Notes: Intent-to-treat estimates; monetary outcomes are winsorized at the 99 percent level and in US dollars. All regressions include strata dummies and include the baseline value of the outcome. Mobile Account is the treatment where only a mobile money account was provided and the loan was disbursed as cash. Mobile Disburse is the treatment where a mobile money account was provided and the loan also disbursed onto this account. “Profit” refers to the woman’s self-reported monthly business profit. “Savings” is individual savings held by the woman. “Capital” is the value of all assets the woman uses in her business plus the value of inventory held for her business. Control mean endline is the mean value of the outcome in the control group at endline. Control mean baseline is the mean value of the outcome in the control group at baseline. False discovery rate (FDR) adjusted p -values, also known as q -values, were used to correct for multiple hypothesis testing. They are shown in curly brackets. These were calculated following the method of Benjamini, Krieger, and Yekutieli (2006). Robust p -values in square brackets. Robust standard errors in parentheses.

business profits or capital despite also receiving a loan, aligned with null results found in Banerjee, Karlan, and Zinman (2015).¹⁶

In online Appendix Figures A3 and A4, I show cumulative distribution functions for the three outcomes by treatment group in logs and values. I see that for both profit and business capital, the Mobile Disbursement treatment group shows a strong shift to the right in the CDF. Using a two-sample Kolmogorov-Smirnov test, I can reject equality of the distributions of profit and capital when comparing the Mobile Disbursement group to control group ($p = 0.002$ and $p = 0.007$ for profit and capital, respectively) and to the Mobile Account group ($p = 0.059$ and $p = 0.030$ for profit and capital, respectively). I find no evidence of differences in the distribution of savings across the treatments.

The Mobile Disbursement treatment allows women to accumulate more assets and inventory in their businesses: women in the Mobile Disbursement treatment have 0.6 additional assets and have increased the total value of their business assets by US\$37 and of their inventory value by US\$33 (online Appendix Table A8).

¹⁶Note, though, that these results are for eight months after the loan was disbursed and cannot capture intermediate uses or impacts. Section VC discusses evidence on how the loan was used immediately after it was received. I do see an increase in total household assets in the control group of US\$200; see online Appendix Table A8 and Section VIIC for a discussion of this.

Hence, businesses that received the Mobile Disbursement treatment seem to be buying both more inventory and one moderate-value asset of a different variety to their existing assets.¹⁷

The results are robust to alternative specifications and the treatment of outliers (online Appendix Tables A11–A13). I perform a permutation test in online Appendix Table A14. The permutation *p*-values reject the null hypotheses at the same levels as the robust *p*-values.¹⁸

B. Mobile Money Transactions

I look at mobile money account usage outcomes based on administrative data collected from the mobile telecoms operator, MTN, to see whether the accounts were mainly used for transactions or storage of the loan.¹⁹

A summary of the mobile money account usage outcome statistics for those who received a SIM card is shown in online Appendix Table A15.²⁰ “Ever deposit” captures if the woman ever deposited money onto the mobile money account, for example, by topping up the account herself, receiving money from someone else, or by being paid for goods or services on the account. It excludes the loan disbursement for the Mobile Disbursement group. As seen in the table, both groups are similarly likely to deposit money onto the account, with 15 percent ever depositing. This means that for the Mobile Account group, only 15 percent ever used the account (since they could not withdraw or save money without first depositing some). Both groups make similar low numbers of deposits (0.6–0.8 of a deposit at the mean), and the deposit amount conditional on making a deposit is similar for both treatments at US\$12–US\$15. The most common outcome for both groups is that they don’t deposit anything to the account.

Larger differences appear between the treatments when looking at withdrawals. The Mobile Disbursement treatment group make a withdrawal 96 percent of the time.²¹ For the Mobile Account group, withdrawals are similar to deposits, at 12 percent ever making one. The number of withdrawals is also much higher for the Mobile Disbursement group. This is important, as in principle, the Mobile Disbursement group could just withdraw all the loan the day they got it and so only needed to make one withdrawal. However, on average, women in the Mobile Disbursement treatment make nearly 4.4 withdrawals. Likewise, the average withdrawal amount for the Mobile Disbursement group was less than the average loan size: US\$180 as an average withdrawal compared to the average loan size of US\$380. Fifty-seven percent of the Mobile Disbursement group made a withdrawal on the same day that

¹⁷ I do not see any impact of either treatment on hours worked by either household or nonhousehold employees or the number of employees (see online Appendix Table A9). I do not see any changes in the type of business operated as a result of the treatment (see online Appendix Table A10).

¹⁸ Since I have multiple treatments and strata in my randomization, I also check for contamination bias, as described in Goldsmith-Pinkham, Hull, and Kolesár (2022). I do not find evidence of contamination bias.

¹⁹ These data also allow me to verify that the loan was successfully disbursed onto the mobile money account for the 674 of the 982 women assigned to Mobile Disbursement. This is 26 fewer than the 700 recorded as receiving the loan on the mobile money account in the survey data.

²⁰ In online Appendix Table A16, I show regression estimates of the same statistics.

²¹ This is not 100 percent, as these statistics include all women who received a SIM card, some of whom did not receive their loan on the SIM card.

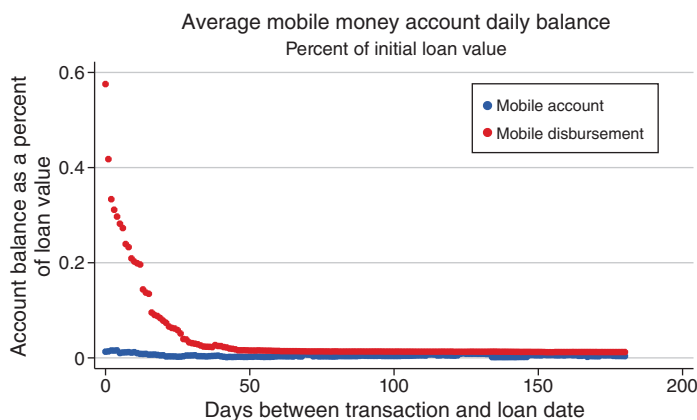


FIGURE 2. MEAN BALANCE IN THE MOBILE MONEY ACCOUNT OVER TIME BY TREATMENT GROUP

Notes: Balance on the account as a percent of the loan disbursed value; balance captured at end of day for a six-month period after the disbursement of the loan. Includes those given a SIM card in the Mobile Account arm and those who had their loan disbursed on the mobile money account for Mobile Disburse; excludes those who didn't receive their assigned treatment.

the loan was disbursed, and this was on average for 42 percent of the loan value. Qualitative questions and survey responses suggest this was not because mobile money agents didn't have enough float to withdraw all the loan at once but because the women were choosing to retain some money on the accounts.

In Figure 2, I show the daily end-of-day balance on the mobile money account as a percentage of the disbursed loan amount by treatment status. If the Mobile Disbursement group simply withdrew the entire balance on day 0, the end-of-day balance would be zero. I do not see this. Instead, the Mobile Disbursement group held large balances on the account in the period immediately after loan disbursement: almost 60 percent of the loan value is kept on the account beyond the first day, and the balance is drawn down slowly over approximately 2 months. The Mobile Account group held near-zero balances throughout the period.

I show regression estimates for the same end-of-day balance outcomes in online Appendix Table A17. During the first 7 days after loan disbursement, women in the Mobile Disbursement group are on average holding US\$120 on the account, approximately 32 percent of the loan value or 100 percent of baseline savings. Between days 8 and 15, this falls to US\$63, and by days 15 and 30, this falls to US\$23. This compares to an average balance of only US\$0.8 held by the Mobile Account group during the first week after loan disbursement, US\$0.5 between days 8 and 15, and US\$0.23 by days 15–30.

The majority of transactions are cash-out (online Appendix Figure A5) (55 percent of transactions for the Mobile Disbursement group, 32 percent for the Mobile Account group), though both treatment groups also often make payments to buy air-time or data (30 percent of transactions for both groups) and make transfers to and from other mobile money accounts (10 percent of transactions for both groups). The mean value of transactions by type by treatment group is shown in online Appendix Table A18. For the Mobile Disbursement group, the cash-out transaction type has a mean value of US\$123 and hence represents the main method by which people

withdrew their loan. Transfers out to another mobile money account are only US\$38 in the Mobile Disbursement group, so while some women are withdrawing their loans by sending it to another mobile money account (which could include paying suppliers with mobile money), this is not the majority. Note that only 2 percent of transactions (10 percent of transfers out) are where a woman sent money to the mobile money account of the phone numbers provided in the survey, so this does not seem to be a frequent type of transaction. Less than 0.5 percent of transactions (3 percent of transfers out) are to the woman's spouse's phone number.

Overall, the summary of transaction records suggests that for both treatments, the mobile money accounts were not used for frequent transactions and hence were not used extensively for either business transactions or savings.²² Instead, it appears the accounts were predominantly used by the Mobile Disbursement group to store the loan and draw it down over time, cashing it out when needed to purchase something for the business and making multiple withdrawals of part of the loan value.

V. Mechanisms: Self-Control, Spousal Pressure, or Saving Constraints

There are three main channels through which disbursement of loans onto mobile money accounts could impact women's businesses: Firstly, it could have facilitated both learning and credibility about saving in a mobile money account and so relaxed saving constraints. Secondly, it may have helped women to exercise self-control, both through mental accounting effects of having an earmarked account for the business and through the soft commitment of having to withdraw money from the account rather than have it as cash in hand. Finally, it may, by keeping the loan in a digital form labeled for the business, have circumvented sharing pressure and so enabled women to invest more of the loan in their businesses.

A. Saving Constraints

One way the mobile money accounts could have an effect is if the women were saving constrained. It is possible that women had not thought to use their mobile money accounts as a saving device, given mobile money is designed for sending remittances. Consistent with this, nationally only 16 percent of mobile money users report using it to save (Intermedia 2016), and, in my sample, only 12 percent of the control group reported saving on a mobile money account. In this case, getting the loan directly disbursed on the mobile money account may have had a larger impact due to learning about the benefits of storing money on the account. They may also have implicitly assumed BRAC was validating that keeping so much money on a mobile account is safe and a good idea, helping them to overcome any reservations about doing this.

However, saving constraints seem unlikely as an explanation for my findings: the Mobile Disbursement group were not more likely than the Mobile Account group to

²²This differs from the findings of other studies of labeled mobile money accounts that offered a high interest rate on the account (Dizon 2017; Bastian et al. 2018; Habyarimana and Jack 2024; Batista and Vicente 2020) but is consistent with De Mel et al. (2022). Somville and Vandewalle (2018) and Field et al. (2021) also both find limited own payments into a saving account, with increases in savings driven by automatic payments into the account.

deposit into their account. I also do not find evidence that those treated with Mobile Disbursement deposited subsequent BRAC loans onto the mobile money account, again suggesting learning and saving constraints cannot explain my findings (see Section VIA for further details on deposit of subsequent loans). Self-reported savings with mobile money, while significantly different for both treatments from the control group, are of economically tiny magnitudes (see online Appendix Table A7—the treatments increase mobile money savings from 2 percent of all savings to 3 percent and 5 percent in the Mobile Account and Mobile Disbursement treatments, respectively). The women also already had access to many other forms of saving, including over one-third who save in a bank account. This makes it doubtful that saving constraints can explain my effects.

B. Self-Control

Having a designated account for the business may help women overcome self-control difficulties through mental accounting effects. The small fees charged to withdraw mobile money may also act as a commitment device limiting small, impulse purchases.

To examine if self-control difficulties are a key channel through which the accounts had an impact, I look at heterogeneity by an index of self-control difficulties at baseline, following Anderson (2008).²³ The index is composed of whether a woman had hyperbolic time preferences (which was stratified on in the randomization); whether she was impatient, where impatience was defined as always preferring money now over the future in the near-far time frame; and whether she didn't report saving for her business (the construction of the index is summarized in online Appendix Table A19).²⁴

As shown in Table 2, I see no heterogeneous effects by the index of prior self-control difficulties for either treatment. This contrasts with Somville and Vandewalle (2018) and Fafchamps et al. (2014), who argue self-control difficulties can explain their findings.

C. Family Pressure

During focus groups, the women discussed the pressure they experienced to share some of the loan with their family when they first get it. This is compounded by the visibility of large amounts of cash in small-denomination bills. The women discussed the many strategies they employ to quickly use all or part of the loan when they first receive it, even spending on household items or their children rather than keep the loan as cash.

When the loan is disbursed onto a mobile money account, it is not subject to the same pressure around sharing cash on hand (Platteau 2000). Default effects may

²³The components of this index were prespecified as heterogeneity in the pre-analysis plan. However, combining them into an index was not prespecified. I do this to improve power. The individual components of the index, alongside all prespecified heterogeneity, can be seen in online Appendix Tables A32–A34. Results for this and other indexes are unchanged using an index constructed using principal components.

²⁴While a component of the self-control index was used to stratify the original randomization, the other components could be picking up a correlation with another variable.

TABLE 2—HETEROGENEOUS TREATMENT EFFECTS BY BASELINE SELF-CONTROL AND FAMILY PRESSURE INDEX

	Profit (1)	Savings (2)	Capital (3)
<i>Mobile Account</i>	−0.50 (6.02) {0.99}	2.60 (14.96) {0.99}	15.37 (37.32) {0.99}
<i>Mobile Disburse</i>	1.15 (5.67) {0.99}	12.85 (16.54) {0.99}	−20.96 (36.38) {0.99}
<i>MA × self-control</i>	−1.43 (8.05) {0.99}	2.01 (20.51) {0.99}	19.42 (51.95) {0.99}
<i>MD × self-control</i>	9.92 (7.55) {0.99}	6.94 (21.32) {0.99}	38.06 (50.22) {0.99}
<i>MA × family pressure</i>	9.04 (7.87) {0.99}	−4.05 (21.52) {0.99}	−24.81 (53.18) {0.99}
<i>MD × family pressure</i>	30.29 (7.63) {0.00}	−15.03 (22.74) {0.93}	183.55 (51.33) {0.00}
<i>Family pressure</i>	−10.78 (5.53)	26.06 (16.52)	8.46 (40.17)
Observations	2,639	2,639	2,639
R^2	0.44	0.35	0.52
Control mean self-control	107.57	141.68	656.28
Control mean family pressure	106.20	174.82	726.44
Control mean baseline self-control	118.10	123.72	658.17
Control mean baseline family pressure	119.31	146.28	701.97
<i>p-val MD self-control = MD family pressure</i>	0.05	0.47	0.04
<i>p-val MA self-control = MA family pressure</i>	0.31	0.84	0.55

Notes: Intent-to-treat estimates; monetary outcomes are winsorized at the 99 percent level and in US dollars. All regressions include strata dummies. Mobile Account (MA) is the treatment where only a mobile money account was provided and the loan was disbursed as cash. Mobile Disburse (MD) is the treatment where a mobile money account was provided and the loan also disbursed onto this account. Heterogeneous indexes are defined in Section V, and the construction is shown in online Appendix Table A19. The interaction is for someone who is above the median in the index. “Profit” is self-reported monthly profit. “Savings” is total savings in each form of saving used. “Capital” is composed of business assets and inventories. False discovery rate adjusted *p*-values, also known as *q*-values, were used to correct for multiple hypothesis testing. They are shown in curly brackets. These were calculated following the method of Benjamini, Krieger, and Yekutieli (2006). Robust standard errors in parentheses.

mean that the loan remains in the form it starts off in, with women in the Mobile Disbursement group keeping part of the loan in a digital form, where it is not subject to sharing pressure in the way cash is. Additionally, though mobile money accounts were designed to send money, they still involve multiple steps to making a transfer, which are considerably more of an obstacle compared to taking cash out of a pocket. A fee must also be paid on all withdrawals, which is a larger percentage of small withdrawals. Direct deposit to the mobile money account by BRAC may also make it more credible for the woman to argue that this money was given to her for her business, and hence earmarked for this purpose, or that BRAC would know if she used it for other things. Both treatment groups could also use the account to obscure

business profits by making deposits to the account, though I do not see this occurring in the data.²⁵

To examine whether mitigating social pressure was the main mechanism by which the mobile money treatments affected women's businesses, I look at an index of family pressure at baseline and examine heterogeneous effects by this index. I construct this index in the same way as for self-control using the method of Anderson (2008). The index is composed of the following components: whether she switched above the median in the control-over-money game (stratified at baseline); whether she was married; whether she reported that when she had money on hand, her spouse and family take it; and whether her spouse or another household member had a business at baseline (the construction of the index is summarized in online Appendix Table A19).

I find strong heterogeneous effects for the Mobile Disbursement treatment by the index of family pressure for both profit and business capital (Table 2).²⁶ Those with high family pressure at baseline see an additional increase in their profits of US\$30 from getting the Mobile Disbursement treatment, or approximately 25 percent of profits at baseline. There is no impact of the Mobile Disbursement treatment for those who didn't experience above-median pressure to share with family at baseline. Those who experienced family pressure to share at baseline see their business capital increase by US\$183, or 26 percent of baseline, from the Mobile Disbursement treatment. The heterogeneous effects by family pressure survive a multiple testing correct for profits and business capital, remaining significant at the 5 percent level. I see no heterogeneous effects from the Mobile Account treatment and no heterogeneous effects for savings.

I also examine whether the withdrawal behavior from the mobile money account differs by sharing pressure in the Mobile Disbursement group. If the mobile money account is assisting women in resisting sharing pressure, I might expect those women who experience the most pressure to leave the loan on the mobile money account for longer. Indeed, I find that women who experienced high pressure to share money with family keep more money on the mobile money account beyond the first 15 days after the loan is disbursed until 180 days after disbursement (online Appendix Table A20).

To further check the robustness of the heterogeneity results, I use the causal forest approach highlighted in Athey and Imbens (2016) and carried out in Davis and Heller (2017) to look at the characteristics of those who experience the greatest and smallest treatment effects from the Mobile Disbursement treatment (see online Appendix Section B). I do this by looking at treatment heterogeneity in four quartiles. I then compare the average baseline characteristics between those in the top quartile (most affected) and those in the bottom quartile (least affected by treatment). This enables me to learn what the characteristics are of those who responded most to the Mobile Disbursement treatment.

²⁵ Since repayment of the loan remained highly visible as cash at community group meetings, and all treatment arms received their loan at the branch, I think it unlikely the Mobile Disbursement treatment is facilitating hiding of the loan or would enable the woman to keep the loan a secret from her family. Women reported that the fact they received BRAC loans would be common knowledge within the community.

²⁶ Heterogeneity by components of this index can be seen in online Appendix Tables A32–A34.

Results are shown in online Appendix Table A21. I see that those who experienced the largest treatment effects on business profit and capital had more profitable and well-capitalized businesses at baseline but were more likely to be married, own their business jointly with their spouse, have another business in their household, self-report that their family takes their money, make fewer decisions in the household, were more willing to hide money from the spouse in the experimental hiding game, and scored significantly higher in the family sharing pressure index. This suggests that women must both experience sharing pressure but also be successful enough to have money to share to benefit from the Mobile Disbursement treatment. Interestingly, it also seems that there is a group of women who saw effects from the Mobile Disbursement treatment on savings and that these women had smaller businesses at baseline and did not experience sharing pressure.²⁷

Spousal or Family Pressure.—In order to examine who in the household is exerting pressure to share money, I look at the individual components of the family pressure index (online Appendix Tables A32–A34). I see that it is primarily willingness to hide money from family, being married, and self-reporting that if you have money on hand, your spouse/family take it at baseline that are driving the index (columns 2, 10, and 13) for profit. The spouse or other family member owning a business also drives heterogeneous effects on business capital (columns 15 and 16), though these estimates are noisier. This suggests that it is primarily the spouse who is the source of the sharing pressure within the household rather than other family members.²⁸

To further validate this, I look at whether the spouse is present in the home or not since presumably there is little sharing pressure from the spouse if he works away from home (online Appendix Table A22). I see that the treatment effect of the Mobile Disbursement is larger if a spouse is present at home.

Expenditure Patterns.—If the Mobile Disbursement treatment helped women to resist family pressure to share money, then this should be apparent in how the loan was used immediately after it was disbursed. Thus, I examine how the loan was used in the week after disbursement. Note that these questions about the use of the loan in the week following disbursement were asked on average eight months later and so may be subject to large measurement error and recall bias compared to other questions, which ask about the current period. They may also be more sensitive for the women to answer since the loan is meant to be explicitly for their business, and so show overreporting of business expenditures. This bias, however, would not be expected to differ by treatment group. I also did not prespecify this outcome in the pre-analysis plan. Nonetheless, it may help us understand how the Mobile Disbursement treatment had an impact on business outcomes.

²⁷I also use clustering analysis to split the sample into different types of women and examine treatment heterogeneity by these clusters (see online Appendix C). I find women who were married, had more successful business that they own themselves, report their family takes their money when they have it, and who hide money in the hiding game are driving the impact of the Mobile Disbursement treatment on profit and capital, confirming that social sharing pressure seems to be the key mechanisms explaining the treatment effect.

²⁸Eighty percent of women live with another adult household member who is not their spouse, mainly adult children, siblings, and nieces/nephews.

TABLE 3—TREATMENT EFFECTS ON USE OF THE LOAN IN THE FIRST WEEK AFTER DISBURSEMENT

	Business (1)	Sharing (2)	School (3)	Home (4)	Expenditure (5)	Saving (6)	Loan (7)
<i>Mobile Account</i>	3.30 (6.60)	1.98 (1.42)	1.32 (1.85)	2.62 (2.94)	−0.06 (0.09)	−2.64 (3.22)	−0.00 (0.09)
<i>Mobile Disburse</i>	4.81 (6.54)	−7.99 (1.36)	−1.30 (1.73)	−8.14 (2.63)	0.04 (0.10)	12.42 (3.40)	0.01 (0.07)
Observations	2,642	2,642	2,642	2,642	2,642	2,642	2,642
R^2	0.20	0.21	0.17	0.16	0.21	0.16	0.11
Control mean	212.34	37.53	31.11	30.80	0.25	42.74	0.12
p -value $MA = MD$	0.82	0.00	0.14	0.00	0.25	0.00	0.89

Notes: Not specified in pre-analysis plan; intent-to-treat estimates. All outcomes are winsorized at the 99 percent level and in US dollars. All regressions include strata dummies. Mobile Account is the treatment where only a mobile money account was provided and the loan was disbursed as cash. Mobile Disburse is the treatment where a mobile money account was provided and the loan also disbursed onto this account. Amount of loan spent on each category one week after receiving loan. “Business” is business inventory and assets. “Sharing” is money given to the spouse, friends, or other family members, both at home and elsewhere. “School” is money spent on school fees and related expenditures. “Home” is money spent on items for the home or home improvements. “Expenditure” is money spent on food, clothes, transport, etc., and “Loan” is money spent paying back other loans. Recall eight months later. Average loan amount was US\$380. Robust standard errors in parentheses.

Results for how the loan was used in the week after it was disbursed across seven categories are shown in Table 3. Spending on the business was the largest use of the loan immediately after disbursement, with an average of US\$212 or 55 percent of the mean loan size of US\$380. However, spending on other categories was also large, with US\$38 going to sharing with others (10 percent), US\$31 on school fees (8 percent), and US\$30 on household assets (8 percent). On average, only US\$42 of the loan is “saved” after the first week, suggesting that the loan is put to use very quickly.

I see significant differences for the Mobile Disbursement treatment in the composition of loan spending. The Mobile Disbursement treatment group spend US\$8 less giving money to their family and US\$8 less on their home. This additional US\$16 is now able to be spent on the business—and indeed, it seems US\$5 was spent immediately on the business. Additionally, US\$12 more was saved than the US\$42 saved beyond the first week after loan disbursement in the control group. Crucially, money saved by the Mobile Disbursement group remains on the mobile money account allocated for the business, whereas money saved by the control and Mobile Account groups as cash is subject to sharing pressure and may not end up being invested in the business. Hence, in the Mobile Disbursement group, it is not just the extra US\$12 they save but the entire US\$54 saved that is available to invest in the business beyond the first week. This also matches the administrative data on the mobile money account balances, where we saw in online Appendix Table A17 that the Mobile Disbursement group had US\$60 stored on the mobile money account on average 1 to 2 weeks after the loan was disbursed. If this entire US\$54 is invested in the business, this can explain most of the US\$70 of additional capital I see in the business of those in the Mobile Disbursement group.

I also examine the woman’s report of how much money she gives her spouse at endline (not prespecified), to gain an idea of how sharing pressure might have changed more generally as a result of Mobile Disbursement (Table 4). I find that

TABLE 4—TREATMENT EFFECTS ON AMOUNT AND WHETHER THE WOMAN GAVE MONEY TO HER SPOUSE AND AMOUNT RECEIVED FROM HER SPOUSE IN THE LAST MONTH

	Amount given spouse (1)	Dummy gave spouse money (2)	Amount received spouse (3)
<i>Mobile Account</i>	−1.16 (1.06)	−0.03 (0.03)	1.15 (2.67)
<i>Mobile Disburse</i>	−2.99 (0.98)	−0.09 (0.03)	−0.51 (2.74)
Observations	1,613	1,613	1,613
R^2	0.24	0.29	0.27
Control mean	6.08	0.30	43.84
Control mean baseline	3.28	0.22	44.46
p -value $MA = MD$	0.07	0.10	0.54

Notes: Not in pre-analysis plan; intent-to-treat estimates. All outcomes are winsorized at the 99 percent level and in 2000 Ugandan shillings. All regressions include strata dummies and include the baseline value of the outcome. Mobile Account is the treatment where only a mobile money account was provided and the loan was disbursed as cash. Mobile Disburse is the treatment where a mobile money account was provided and the loan also disbursed onto this account. Amount given/received spouse is the monthly transfer to/from the spouse. Robust standard errors in parentheses.

women who received the Mobile Disbursement treatment give significantly less money to their spouse each month, US\$3, on a mean of US\$6, or 50 percent less. They are also significantly less likely to give any money to their spouse, with the Mobile Disbursement treatment group being 9 percentage points less likely to give money to their spouse each month. This is on a mean of one-third of women giving any money to their spouse. Interestingly, between baseline and endline, the control group went from giving US\$3.3 to US\$6 and from 22 percent of them giving money to 30 percent of them giving money to their spouse. For the Mobile Disbursement group, there is no change in the probability of giving or the amount given to their spouse. I find no significant impact of the Mobile Account treatment on money given to the spouse or the probability of giving money to the spouse, though the coefficients are negative and I cannot reject equality with the Mobile Disbursement treatment at the 10 percent level.

These findings suggest that following receipt of the loan in the control group, spouses receive higher amounts of money from their wives every month. This may be because receiving a cash loan results in the woman being perceived as cash rich, making her more subject to sharing pressure going forward. The Mobile Disbursement treatment mitigates this increase in sharing. While only US\$3 less a month, if the woman had been giving US\$3 less to her spouse every month for the last 8 months, this reduction in sharing would add up in total to US\$24 more available for the business.²⁹ When combined with the additional US\$54 saved from the loan in the week after it was disbursed in the Mobile Disbursement arm, these more than add up to the additional US\$70 that I see in the business in the Mobile

²⁹This is a strong assumption, and it's perfectly possible that in the intermediate period, the amount of money the woman gave her spouse could have been greater or less than that seen at eight months.

Disbursement group and so can explain how the Mobile Disbursement treatment is enabling more business investment.

I also confirm that those treated with Mobile Disbursement don't receive less money from their spouses. This is shown in Table 4, column 3. While women only give US\$6 to their spouse, they receive on average US\$44 from their spouses. This is unchanged between baseline and endline and does not differ by treatment.

This evidence on heterogeneity, use of the loan, and money given to the spouse is further supported by focus groups. A common theme that ran through all the discussions was the control that the Mobile Disbursement treatment gave to women to use the loan in the way they intended rather than spending it on other things or giving it to other people. Women described the disbursement of the loan onto the mobile money accounts as helping them to refuse requests for money by arguing that "BRAC gave me this money for my business and placed it in this account so that I would only use it for my business. If I give some to you they'll [BRAC] will know."³⁰

Taken together, this suggests that the Mobile Disbursement treatment reduced sharing pressure through a combination of labeling of the loan as for the woman's business (mental accounting), combined with small transaction costs of accessing the money (soft commitment) plus default effects resulting in money not spent remaining on the mobile money card and earmarked for the business, which together effectively reduced requests for money from family.

Given the control group also got a loan as cash but see no changes in business outcomes, what are they doing with the loan? I show in online Appendix Table A8 (column 6) the impact of the treatments on total household wealth, which is the value of household durable goods not used in the woman's business. Total household wealth increases between baseline and endline for all groups by half (US\$200), which is almost two-thirds of the loan value. This suggests that the control and Mobile Account groups are spending a large portion of their loan on household durables, and hence, it is not surprising that they do not see growth in their businesses.

VI. Why Didn't the Mobile Account Treatment Imitate the Mobile Disbursement Treatment?

One outstanding puzzle is why the Mobile Account treatment did not just imitate the treatment received by the Mobile Disbursement group by depositing their own loan onto the mobile money account they were given. Equally, why didn't the control group, 97 percent of whom already had a mobile money account, also deposit some of their loan onto the account?³¹

³⁰BRAC never had access to the account transaction data; only the researcher did, and the women were informed of this at the start of the study. The woman saying this in the focus group knew BRAC didn't actually have the ability to know what she used the loan for if it came out of the mobile money account but seemed to be using the fact that other people didn't know this to refuse their requests for money.

³¹Note that while imitation of the Mobile Disbursement treatment was entirely possible by the Mobile Account and even control groups, since I provided a small amount for withdrawal fees to the Mobile Disbursement group, the benefits of imitating would not be as large.

A. Learning

Only 12 percent of the sample saved with mobile money at baseline. There may therefore have been learning effects around keeping so much money on a mobile money account or from BRAC legitimizing storing funds on a mobile money account. Additionally, women may have needed to experience the benefits of mobile money deposit of their loans.

If learning about the benefits of mobile disbursement took place, women in this treatment arm would prefer to receive subsequent loans on the mobile money account. I look at this in Table 5, columns 1 and 2: 67 percent of the control group would like to receive their next loan as mobile money, while a significantly higher 71 percent of the Mobile Account and 81 percent of the Mobile Disbursement group would. I see that the entire increase in both treatment arms' preference for a loan to be disbursed on a mobile money account is driven by those women who were above the median in the sharing pressure index, suggesting women learned of the benefits of receiving a loan on a mobile money account when they experienced high sharing pressure.³²

To test whether this learning translates into changes in behavior, I examine whether the treatment groups chose to deposit any subsequent loans from BRAC onto the mobile money account. I do this by combining BRAC admin data with the mobile money transaction records for the whole of 2017. I restrict the sample to women taking out a subsequent loan during 2017, of which 1,417, or almost half my sample, did, and examine the deposits and withdrawals to the mobile money account I provided. I do not see any evidence that women in the Mobile Disbursement group deposit subsequent loans onto the mobile money account or evidence of heterogeneity by sharing pressure (Table 5, columns 3 to 8).³³ Only 12 percent of the Mobile Account and Mobile Disbursement groups ever make a deposit to the mobile money account in the two weeks after the subsequent loan was disbursed, and the amounts deposited are extremely small (US\$3 on average). As a share of the loan value, this is less than 1.5 percent. Further, I do not see any evidence that the Mobile Account group start to deposit their loan onto the mobile money account over time, as they observe other members of their group receive the loan on the mobile money account (online Appendix Figure A6).

It seems that something other than learning is preventing women from depositing the loan themselves onto a mobile money account. In other words, there is a difference between making the deposit yourself rather than BRAC directly depositing the loan on the mobile money account. This could be a real cost in terms of additional time or money to do the deposit yourself or a mental difference in how the money is perceived based on who makes the deposit.

³²It is still possible that women are underestimating the benefits of depositing the loan on the mobile money account.

³³It is possible that women are instead using their personal mobile money accounts to deposit the loan, in which case I would not be able to observe this.

TABLE 5—TREATMENT EFFECTS ON STATED AND ACTUAL USE OF THE MOBILE MONEY ACCOUNT FOR SUBSEQUENT LOANS

	Stated preferences		Own deposit behavior					
	Want mobile deposit (1)	Want mobile deposit (2)	Any own deposit (3)	Any own deposit (4)	Own deposit amount (5)	Own deposit amount (6)	Own deposit share (7)	Own deposit share (8)
<i>Mobile Account</i>	0.04 (0.02)	−0.00 (0.03)						
<i>Mobile Disburse</i>	0.14 (0.02)	0.10 (0.03)	−0.03 (0.03)	−0.03 (0.04)	−1.21 (1.02)	−1.47 (1.45)	−0.01 (0.01)	−0.00 (0.01)
<i>MA × family pressure</i>		0.10 (0.05)						
<i>MD × family pressure</i>		0.10 (0.04)		−0.01 (0.05)		0.54 (1.90)		−0.01 (0.01)
Observations	2,639	2,639	964	964	964	964	964	964
R^2	0.21	0.22	0.35	0.35	0.78	0.78	0.85	0.85
Control mean	0.67	0.68	0.12	0.11	2.81	2.78	0.01	0.02
p -value $MA = MD$	0.00	0.00						

Notes: Intent-to-treat estimates; all regressions include strata dummies. Stated preferences refer to the responses women gave when asked hypothetical questions about receiving future loans on a mobile money account. “Want mobile deposit” means that the woman indicated in the endline survey that she would like to receive a subsequent loan on a mobile money account. “Own deposit” variables capture if the woman made any deposit of her subsequent loan herself to the mobile money account. A subsequent loan is any loan disbursed in 2017 after the loan that disbursement was randomized for in this study. This is only examined within the treatment groups. “Any own deposit” means a deposit was made to the mobile money account in the two-week period after the subsequent loan was disbursed. “Own deposit amount” is the maximum single deposited amount in the two weeks after the subsequent loan was disbursed, in US dollars. “Own deposit share” is the value of the deposited amount as a share of the subsequent loan amount. Mobile Account is the treatment where only a mobile money account was provided and the loan was disbursed as cash. Mobile Disburse is the treatment where a mobile money account was provided and the loan also disbursed onto this account. Family pressure is a dummy variable if the woman experienced sharing pressure from her family that was above the median at baseline. Control mean is the mean value of that outcome in the control group for stated preferences and the Mobile Account treatment group for own deposit behavior. Robust standard errors in parentheses.

B. Psychological Differences of Direct Deposit

It is possible that going to an agent yourself and depositing some of the loan would not sufficiently earmark the loan for the business compared to BRAC depositing the money for you. It may further be easier to argue that BRAC would “know” the loan was not being used for the intended purpose when BRAC deposits the loan on the digital account themselves. If this is the case, then women can only overcome this through BRAC depositing the loan for them, not through their own actions.

However, given that the main person the woman is getting pressure from to share the loan with is the spouse, and he does not know whether BRAC deposited the loan for the woman or gave her cash that she deposited onto the mobile money account, this doesn’t entirely explain the lack of deposit of the loan by women. Additionally, women seem to engage in widespread hiding of money from their spouses, and from the focus group discussions, this did not seem to cause guilt. Given the large number of mobile money agents available in Kampala, it seems simple for the woman to go directly from BRAC to an agent and deposit her loan, without her spouse knowing

about it.³⁴ However, when discussed during the focus groups, women highlighted that they would struggle with hiding this if “called out” on the behavior. Being found out could also open women up to potential backlash. It is therefore possible that there is a psychological difference between having BRAC deposit the loan for you on the mobile money account and doing this yourself, such that the two are not equivalent.

C. Procrastination

A third hypothesis relates to the time investment in depositing the loan into the mobile money account. Evidence has shown that even small costs can have significant impacts on behavior, particularly for those with hyperbolic preferences (O’Donoghue and Rabin 1999). While women are on average only 5 minutes from a mobile money agent, even this cost given the higher rates of present bias (34 percent) and impatience (20 percent) may have been enough of a deterrent to the women to prevent them from depositing the loan themselves.³⁵

D. Default Effects

A final explanation is default effects. Default effects have been shown to have large impacts on behavior, particularly saving behavior (Choi et al. 2004; Chetty et al. 2014a). Prior studies have demonstrated powerful default effects by showing that money deposited directly into a bank account is saved in the account, whereas cash given at the bank is not deposited onto the account, resulting in less saving (Brune et al. 2017; Somville and Vandewalle 2018). Likewise, employees save more when money is automatically deducted from their wages despite almost no cost of doing it themselves (Blumenstock, Callen, and Ghani 2018; Brune, Chyn, and Kerwin 2021). In my study, Mobile Disbursement makes the default that money remains on the account until actively withdrawn. It is therefore possible that the lack of imitation of Mobile Disbursement by those assigned to Mobile Account is due to default effects and the inertia associated with them, potentially combined with some small cost of depositing money oneself, underestimation of the benefits of doing so, and procrastination to avoid this cost.

Overall, it appears that while women did learn about the benefits of having their loan deposited on a mobile money account, particularly those who experience sharing pressure, an additional barrier prevented them from making the deposit themselves. While it is possible there is a psychological difference that makes it less credible to do the deposit yourself, this seems unlikely as the full explanation given the widespread hiding that women engage in. Instead, it seems like default effects combined with small costs of making the transaction yourself explain why the Mobile Account group did not imitate the Mobile Disbursement treatment.

³⁴The fact that women do not deposit the loan on the mobile money themselves is consistent with other studies that have shown differences between in-kind and cash grants (Fafchamps et al. 2014).

³⁵I also do not see heterogeneity in the impact of the Mobile Account treatment by distance to the nearest agent (see online Appendix Tables A32–A34).

TABLE 6—TREATMENT EFFECTS ON HOUSEHOLD OUTCOMES

	Total household income (1)	Total household consumption (2)	Food consumption (3)	Nonfood consumption excl school (4)	School expenditure (5)
<i>Mobile Account</i>	2.79 (9.90)	7.55 (6.66)	2.67 (2.73)	2.19 (2.36)	3.45 (3.54)
<i>Mobile Disburse</i>	24.21 (10.13)	14.07 (6.74)	5.69 (2.86)	1.35 (2.35)	6.13 (3.34)
Observations	2,642	2,642	2,642	2,642	2,642
R^2	0.33	0.34	0.24	0.20	0.39
Control mean endline	280.48	270.44	112.77	70.13	83.50
Control mean baseline	289.23	246.29	110.65	62.31	70.19
p -value $MA = MD$	0.03	0.33	0.29	0.73	0.43

Notes: Intent-to-treat estimates; all outcomes are winsorized at the 99 percent level and in US dollars. All regressions include strata dummies and include the baseline value of the outcome. Mobile Account is the treatment where only a mobile money account was provided and the loan was disbursed as cash. Mobile Disburse is the treatment where a mobile money account was provided and the loan also disbursed onto this account. All values are monthly for the entire household. Nonfood consumption excludes temptation spending and transfers. Control mean endline is the mean value of the outcome in the control group at endline. Control mean baseline is the mean value of the outcome in the control group at baseline. Robust standard errors in parentheses.

VII. Alternative Explanations

I examine in turn a number of different potential reasons for the results I find, including redistribution within the household, backlash effects against the woman, experimenter demand effects, impacts on tracking transactions, changes in sharing, and worsening loan repayment.

A. Redistribution within the Household

Improving women's ability to invest the loan in their business could reduce household income if other household members' businesses have a higher return to capital (Bernhardt et al. 2019). I look at total household income and household consumption to check for this in Table 6.³⁶ I see an increase in household income of US\$24 in the Mobile Disbursement group. There are no differences in either the spouse's or other household members' business earnings and a small (insignificant) increase in their wage earnings (online Appendix Table A25). I also see significant increases in overall consumption for the Mobile Disbursement treatment, driven by increases in food expenditure and schooling expenditure, which aligns with spending categories that women have shown a preference for (Attanasio and Lechene 2014).

These results contrast with those found in Bernhardt et al. (2018). This could be because women in this sample are willing to pay a high amount to control money and may be engaging in costly strategies to retain control over their loans.³⁷ If the

³⁶Note that the incomes of other household members are as reported by the woman, they were not asked directly, and hence, if the husband keeps some of his income hidden from the woman, I may not be able to observe household income effects.

³⁷Over 55 percent of the sample would be willing to give up US\$8 to retain control of money over giving it to their spouse.

mobile money disbursement of the loan alleviates costly strategies by providing a more effective storage device that the woman can maintain control over, then more profit and overall household income can be generated from the loan.

B. Backlash and Female Empowerment

Giving the woman more control over her loan may have resulted in a backlash against her by her spouse since he now gets less money from her (as seen in Table 4). I did not ask direct measures of discord, arguments, or violence in the household. However, I see no differences in happiness or life satisfaction by treatment group and a small decrease in worries about money for the Mobile Disbursement group (online Appendix Table A26). I also see in online Appendix Table A27 an increase in decisions made by the woman alone in the Mobile Disbursement treatment group (column 2) and an increase in an index of female empowerment (columns 7 and 8). Overall, these results suggest that women are not experiencing a backlash from their spouses.

C. Experimenter Demand Effects

The salience of giving mobile money accounts designed for the business and of disbursing the loans specifically onto those accounts may have caused those women who received the Mobile Disbursement treatment to overreport their business outcomes because they believed that is what the study intended to do. However, this is an unlikely explanation for my findings. First, I also find an increase in consumption, which women would have little incentive to inflate (Table 6). Second, I would expect experimenter demand effects to also influence the Mobile Account group, but I see no impacts of treatment on most outcomes for them.

D. Tracking Transactions

Improved tracking of transactions and record keeping are alternative ways that the mobile money accounts could have improved business outcomes. However, these are unlikely to explain the impacts I see for the following reasons.

Firstly, the mobile money accounts were not used by the majority of the women for frequent deposits and withdrawals of funds. The treatments therefore are unlikely to have made it easier to keep track of regular business expenses and sales since these activities did not take place on the accounts. Additionally, I would only see the treatments correcting measurement error if measurement error only downwardly biased estimates of profit and business capital. It is not clear why measurement error would only downwardly bias reported business outcomes.

Secondly, the balance on the accounts in the Mobile Disbursement treatment was withdrawn to near 0 by 90 days after loan disbursement. If the treatment allowed better tracking of business expenditures, I would not expect to still find impacts eight months later when the accounts are barely used anymore. However, I cannot rule out that getting the loan on the mobile money account made it easier to keep track of the loan, and that resulted in more of the loan being invested in the business rather than spent on other things or given to other people.

Lastly, to try and see if the Mobile Disbursement treatment led to any permanent changes in record keeping, I look at a variable capturing the sort of record keeping occurring in that business. In online Appendix Table A28, I see that there is no impact of treatment on the likelihood of using any method of record keeping, including, importantly, electronic record keeping on a mobile phone, which less than 1 percent of the sample in any treatment arm reports using to keep records.

E. Sharing

If women are giving less to others, they may also receive less support and be less able to withstand shocks. I check for changes in the woman's social support in two ways.³⁸ Firstly, I look at remittance flows to people outside the household in online Appendix Table A29. I see small coefficients on the amount of money sent for both the Mobile Account and Mobile Disbursement treatments of approximately US\$3. I see no other large or significant effects of the treatments on other remittance outcomes. This suggests women are not being cut off from their wider remittance networks. Secondly, I look at women's peer support in the microfinance group in online Appendix Table A30. On average, out of 21 group members, women talk to 7 of them at least once a week outside the group but would ask for help from, and be happy to give help to, just 4 of these. I find no difference in interaction with group members by treatment status, suggesting getting the loan on a mobile money account did not isolate women from other members of their group.

F. Changes in Loan Default

Finally, I check whether the treatments lead to any changes in loan performance. I use BRAC administrative data to look at whether those receiving their loan on a mobile money account were more likely to miss a payment, how late any missed payments were, the balance due on the principle and interest, the balance due of any missed payments and whether there are any changes in the saving balance they hold with BRAC (online Appendix Table A31). I do not see any changes in repayment behavior as a result of my treatments, with default remaining extremely rare.

VIII. Conclusion

This paper shows that the manner in which loans are disbursed to microfinance clients leads to significant differences in how those loans are used. Women randomly assigned to receive the loan on a mobile money account stored a significant balance on the account that they drew down over a number of months. They invested in 11 percent more business capital, had 15 percent higher monthly profits, and had higher household consumption. Impacts were largest for women who experienced high family pressure to share money. This suggests the benefits to women's businesses from the Mobile Disbursement treatment came from a way to store the loan

³⁸I already saw in Table 4 that women receive the same support from their spouses.

that's not subject to the same sharing pressure as cash and ready to invest in the woman's business when needed.

My study suggests that microfinance loan providers should consider directly disbursing loans onto a private account, as opposed to the current default in much of the world of cash. This small change could have significant benefits for the profitability of female entrepreneurs' businesses, at a low cost. Women also had strong demand for the digital disbursement of loans. While my study looked at mobile money accounts due to the fact that they are a widespread and well-understood technology in Uganda, any private account that the woman could control and access easily, such as a bank account in countries where they are common, could have similar benefits. Further research would be needed to understand whether my findings would generalize to contexts where financial services were more developed, transfers cheaper, or payments for goods and services with digital methods common.

A limitation of my study is the relatively short follow-up period of only eight months. BRAC reverted to disbursing loans using only cash after the study ended, preventing an examination of the effects when multiple loans are received in a digital manner.³⁹ Additionally, I only interviewed the woman, meaning that I cannot completely rule out negative impacts on the spouse that the woman was not aware of. Finally, my study only included women in an urban center, but women in rural areas may stand to gain more from increased use of mobile money if they also lack alternative saving accounts. They may, however, struggle with agent access, resulting in them cashing out the loan in one go, and so fail to benefit from the earmarked storage. Future work could study the effects on rural as well as urban borrowers, interview women's spouses, and also look at how the effects persisted over a longer period of time of making loan disbursements using mobile money.

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³⁹BRAC Uganda has recently transformed to a full banking license and has begun the disbursement and collection of microfinance loans using mobile money.

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